Problem Set 3

Erica Lundt

November 12, 2026

Instructions

- Please show your work! You may lose points by simply writing in the answer. If the problem requires you to execute commands in `R`, please include the code you used to get your answers. Please also include the `.R` file that contains your code. If you are not sure if work needs to be shown for a particular problem, please ask.
- Your homework should be submitted electronically on GitHub.
- This problem set is due before 23:59 on Thursday November 12, 2026. No late assignments will be accepted.

In this problem set, you will run several regressions and create an add variable plot (see the lecture slides) in `R` using the `incumbents_subset.csv` dataset. Include all of your code.

Question 1

We are interested in knowing how the difference in campaign spending between incumbent and challenger affects the incumbent's vote share.

1. Run a regression where the outcome variable is `voteshare` and the explanatory variable is `difflog`.

```
reg1 <- lm(voteshare ~ difflog, data = inc.sub)
summary(reg1)
```

   EXPLANATION OF REGRESSION: Holding all things constant, on average for every unit increase in difference in campaign spending, the incumbent and challenger's vote share increases by .042.

2. Make a scatterplot of the two variables and add the regression line.

```r
plot(inc.sub$difflog, inc.sub$voteshare,
     xlab = "Spending Difference",
     ylab = "Inumbent voter share",
     main = "Incumbent vote share vs. Spending differene",
     pch = 16, col = "gray")
abline(reg1, col = "red",lwd = 2)
```

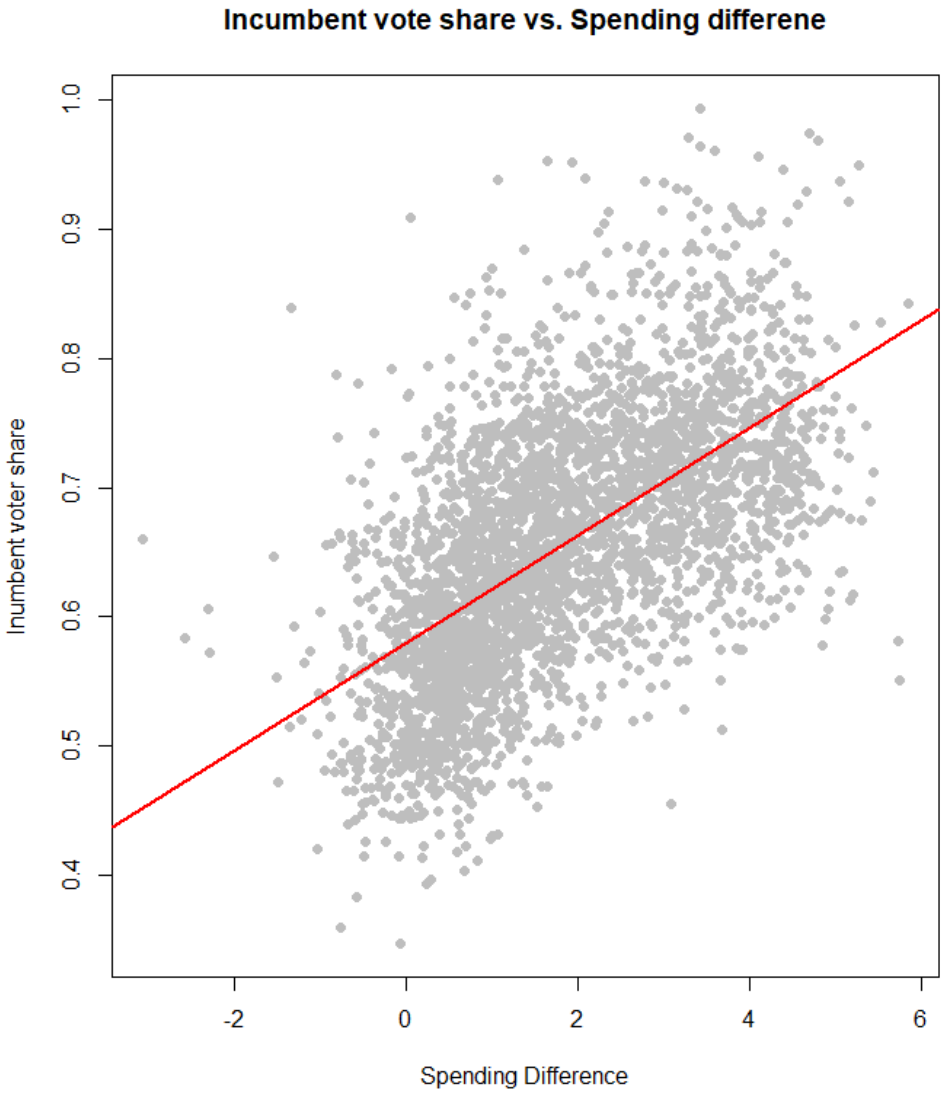


Figure 1: Incumbent Vote Share vs Spending.

3. Save the residuals of the model in a separate object.

```r
# Question 1.c - Save the residuals of the model.
res1 <- residuals(reg1)
```

```
summary(res1)
stargazer(data.frame(Residuals = res1),
          type = "latex",
          title = "Summary of Residuals",
          digits = 4,
          out = "residuals_table.tex")
```

Table 1: Summary of Residuals

Statistic	N	Mean	St. Dev.	Min	Max
Residuals	3,193	−0.0000	0.0787	−0.2683	0.3275

4. Write the prediction equation.

   ANSWER: voteshare = 0.58 + 0.042 * difflog

Question 2

We are interested in knowing how the difference between incumbent and challenger's spending and the vote share of the presidential candidate of the incumbent's party are related.

1. Run a regression where the outcome variable is `presvote` and the explanatory variable is `difflog`.

```
reg2 <- lm(presvote ~ difflog, data = inc.sub)
summary(reg2)
```

   EXPLANATION OF REGRESSION: Holding all things constant, on average for every unit increase in difference in campaign spending, the presidential candidate vote share for the incumbent's party increases by .023.

2. Make a scatterplot of the two variables and add the regression line.

```
plot(inc.sub$difflog, inc.sub$presvote,
     xlab = "Spending Difference",
     ylab = "Pres.Incumbent",
     main = "Incumbent vote share vs. Pres.Incumbent",
     pch = 16, col = "gray")
abline(reg2, col = "red",lwd = 2)
```

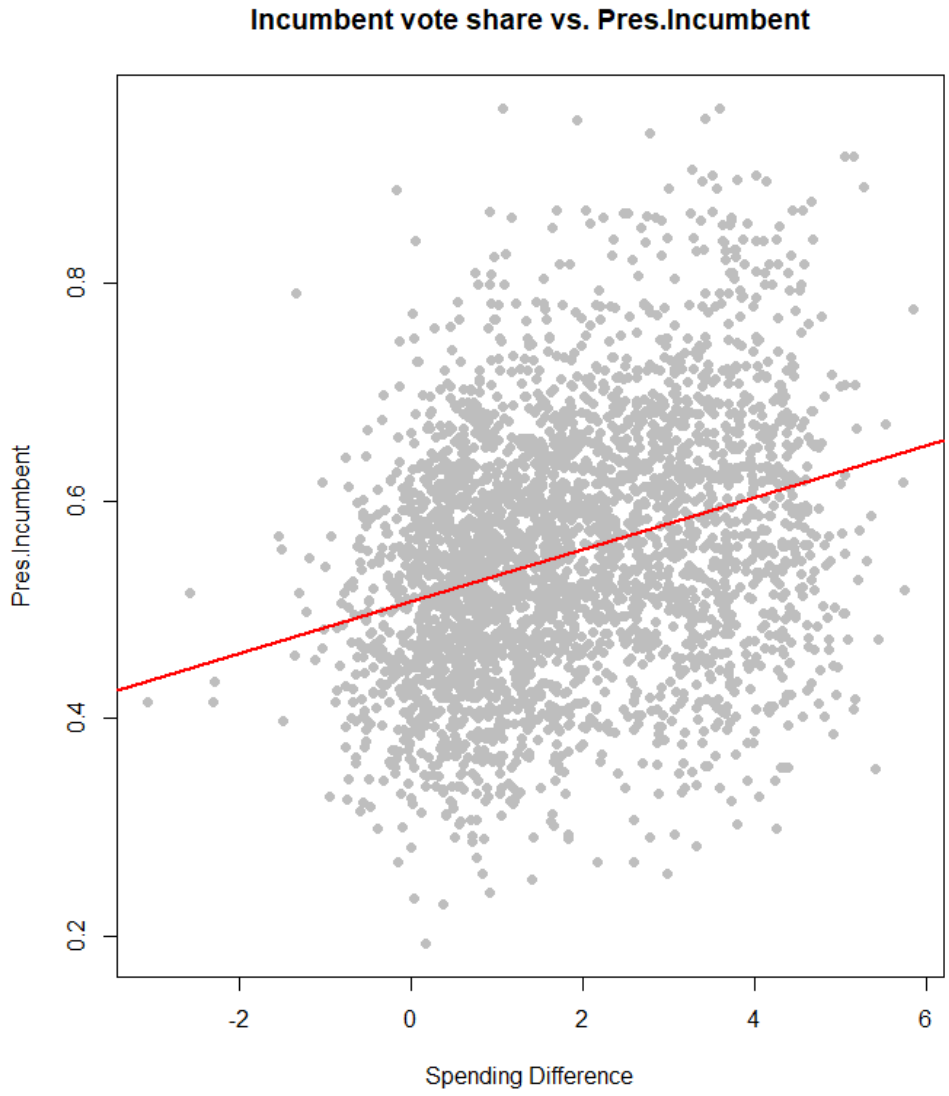


Figure 2: President Incumbent Party vs Spending.

3. Save the residuals of the model in a separate object.

Table 2: Summary of Residuals

Statistic	N	Mean	St. Dev.	Min	Max
Residuals	3,193	0.0000	0.1104	−0.3220	0.4274

```
res2 <- residuals(reg2)
```

```
summary(res2)
stargazer(data.frame(Residuals = res2),
          type = "latex",
          title = "Summary of Residuals",
          digits = 4,
          out = "residuals_table2.tex")
```

4. Write the prediction equation.

   ANSWER: voteshare = -0.21 + .023 * presvote

Question 3

We are interested in knowing how the vote share of the presidential candidate of the incumbent's party is associated with the incumbent's electoral success.

1. Run a regression where the outcome variable is `voteshare` and the explanatory variable is `presvote`.

```
reg3 <- lm(voteshare ~ presvote, data = inc.sub)
summary(reg3)
```

   EXPLANATION OF REGRESSION: Holding all things constant, on average for every unit increase of the vote share, the presidential candidate vote share for the incumbent's party increases by .38

2. Make a scatterplot of the two variables and add the regression line.

```
plot(inc.sub$voteshare, inc.sub$presvote,
     xlab = "Spending Difference",
     ylab = "Pres.Incumbent",
     main = "Incumbent vote share vs. Pres.Incumbent",
     pch = 16, col = "gray")
abline(reg3, col = "red",lwd = 2)
```

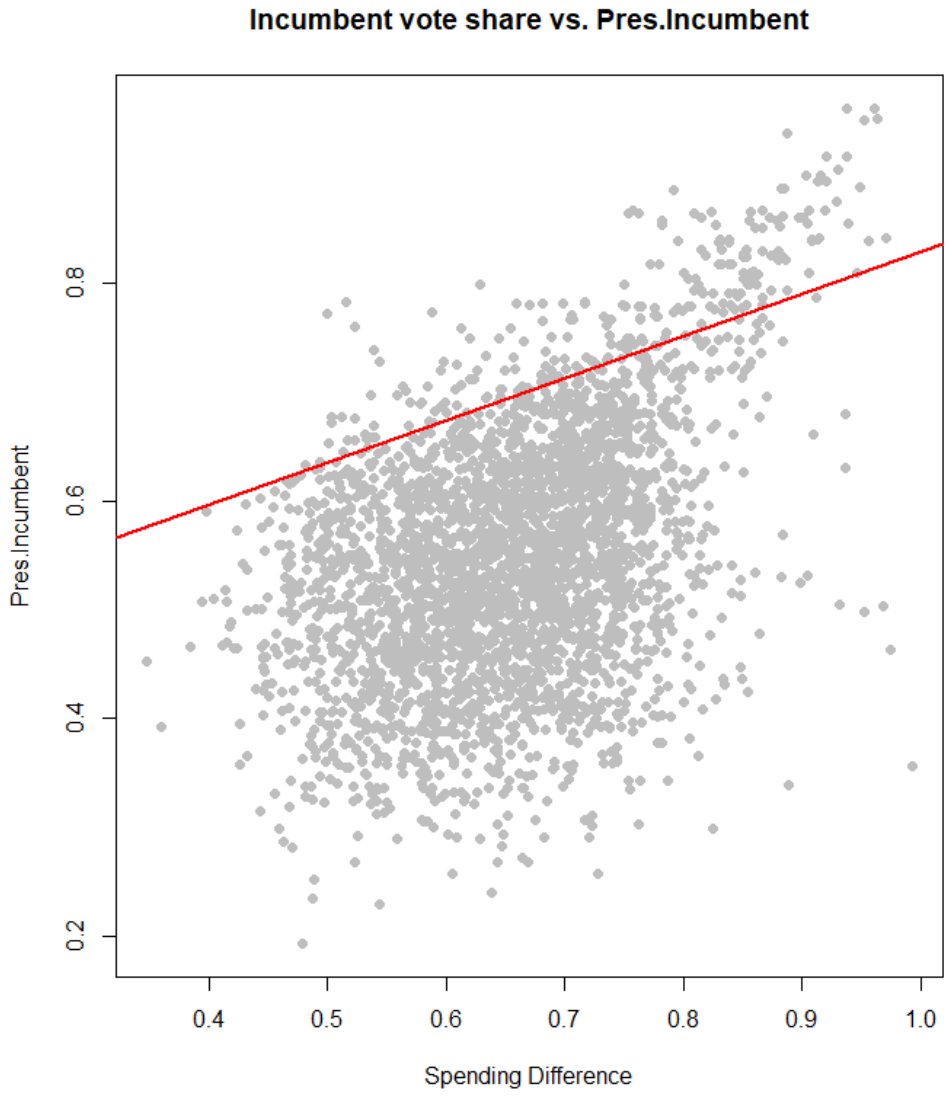


Figure 3: Incumbent Vote Share vs President Incumbent Party.

3. Write the prediction equation.

   ANSWER: voteshare = .044 + .38 * voteshare

Question 4

The residuals from part (a) tell us how much of the variation in `voteshare` is *not* explained by the difference in spending between incumbent and challenger. The residuals in part (b) tell us how much of the variation in `presvote` is *not* explained by the difference in spending between incumbent and challenger in the district.

1. Run a regression where the outcome variable is the residuals from Question 1 and the explanatory variable is the residuals from Question 2.

```
reg4 <- lm(res1 ~ res2)
summary(reg4)
```

2. Make a scatterplot of the two residuals and add the regression line.

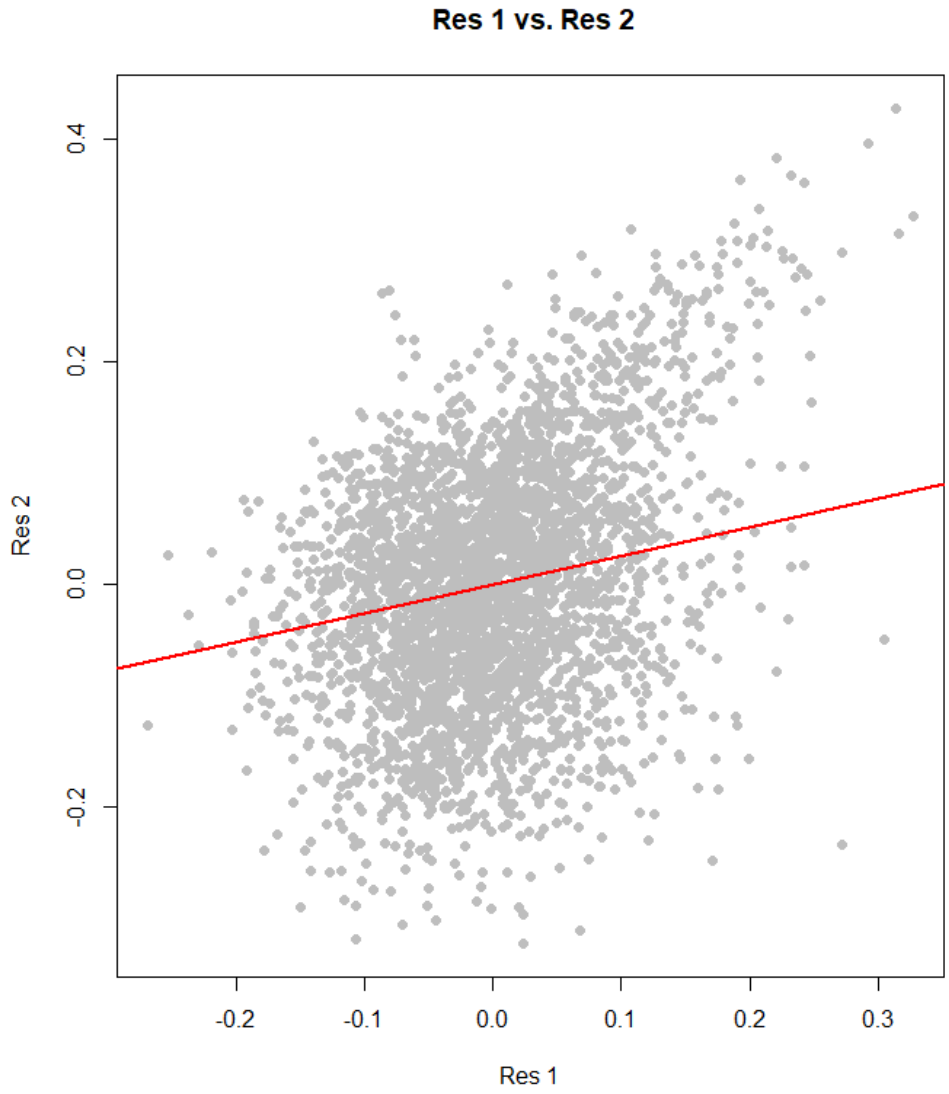


Figure 4: Res 1 vs Res2

3. Write the prediction equation. **ANSWER**: voteshare = -6.5 + -6.1 X res 1

Question 5

What if the incumbent's vote share is affected by both the president's popularity and the difference in spending between incumbent and challenger?

1. Run a regression where the outcome variable is the incumbent's `voteshare` and the explanatory variables are `difflog` and `presvote`.

```
reg5 <-lm(voteshare ~ difflog + presvote, data = inc.sub)
summary(reg5)
```

2. Write the prediction equation. **ANSWER**: voteshare = .44 +.035 X difflog + .25 X presvote

3. What is it in this output that is identical to the output in Question 4? Why do you think this is the case? **ANSWER**:The residuals for the outputs of questions 4 and 5 are identical. I believe this is because regressing one set of residuals on another isolates the variation that remains unexplained by the other variables. When additional variables are added to the regression, they account for that remaining variation, so the residuals end up the same